

Intak Hwang

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I'm a Ph.D. student in [Cryptography & Privacy Lab](#) at Seoul National University, advised by [Yongsoo Song](#). I'm interested in post-quantum cryptographic protocols based on lattices, including but not limited to Fully Homomorphic Encryption and Zero-Knowledge Proofs.

Education

Seoul National University

2023 — 2028

Integrated M.S. & Ph.D. in Computer Science & Engineering

Advised by Yongsoo Song

DGIST

2018 — 2022

B.S. in School of Undergraduate Studies

Summa Cum Laude

Conference Publications

[C09] [2025/1804](#)

HERDS: Multi-Key Fully Homomorphic Encryption with Sublinear Bootstrapping

with Binwu Xiang, Seonhong Min, Zhiwei Wang, Haoqi He, Yuanju Wei, Kang Yang, Jiang Zhang, Yi Deng, Yu Yu

Eurocrypt 2026

[C08] [2024/2032](#)

Carousel: Fully Homomorphic Encryption with Bootstrapping over Automorphism Group

with Seonhong Min, Yongoo Song

Asiacrypt 2025

[C07] [2024/382](#)

On the Security and Privacy of CKKS-based Homomorphic Evaluation Protocols

with Seonhong Min, Jinyeong Seo, Yongoo Song

Asiacrypt 2025

[C06] [2025/216](#)

Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation

with Jinyeong Seo, Seonhong Min, Yongsoo Song

ACM CCS 2025

[C05] [2024/1879](#)

Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption

with Hyeonbum Lee, Jinyeong Seo, Yongsoo Song

ACM CCS 2025

[C04] [2025/1255](#)

Efficient Full Domain Functional Bootstrapping from Recursive LUT Decomposition

with Shinwon Lee, Seonhong Min, Yongsoo Song

SAC 2025

[C03] [2024/1502](#)

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing

with Hyunho Cha, Seonhong Min, Jinyeong Seo, Yongsoo Song

IEEE S&P 2025

[C02] [2024/306](#)

Concretely Efficient Lattice-based Polynomial Commitment from Standard Assumptions

with Jinyeong Seo, Yongsoo Song

Crypto 2024

[C01] [2023/1328](#)

Optimizing HE via Level-aware Key-switching

with Jinyeong Seo, Yongsoo Song

WAHC 2023

Journal Publications

[J01]

A Privacy-Preserving HLA Imputation Method with Homomorphic Encryption

with Hakin Kim, Yongsoo Song, Buhm Han

iScience

Preprints

[P03] [2026/044](#)

Jindo: Practical Lattice-Based Polynomial Commitments for Client-Side Proving

with Hyeonbum Lee, Jinyeong Seo, Yongsoo Song

[P02] [2025/395](#)

Provably Secure Approximate Computation Protocols from CKKS

with Yisol Hwang, Miran Kim, Dongwon Lee, Yongsoo Song

[P01] [2025/203](#)

Ciphertext-Simulatable HE from BFV with Randomized Evaluation

with Seonhong Min, Yongsoo Song

Presentations

Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption

FHE.org 2026 — Taipei, Taiwan

Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation

ACM CCS 2025 — Taipei, Taiwan

FHE.org — [Online](#)

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing

IEEE S&P — San Fransisco, USA

Optimizing HE via Level-aware Key-switching

WAHC 2023 — Copenhagen, Denmark

Projects

TFHE-go

TFHE-go is an implementation of (MK)TFHE scheme, written in Go and Go Assembly. Currently, it is one of the fastest and most feature-complete TFHE implementaion available open-source.

Ringo-SNARK

Ringo-SNARK is a Zero-Knowledge PIOP toolkit for efficiently proving Ring-LWE relations, written in Go. It supports simple, gnark-like circuit design and compilation.

Honors and Scholarships

National Cryptographic Contest

Excellence Award, Encouragement Award	2025
Best Award, Excellence Award	2024
Special Award	2023

CTF Security Competitions

	2020 — 2022
SSTF Hacker's Playground 2022	5th place
WhiteHat Contest 2021	3rd place
DEFCON CTF 2021	Finalist
PlaidCTF 2021	5th place
Real World CTF 2020/2021	1st place
Midnight Sun CTF 2020 Finals	7th place
TokyoWesterns CTF 2020 Finals	3rd place
DEFCON CTF 2020	Finalist

DGIST Dean's List	2020
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Skills

Languages

Korean (native), English (fluent)

Programming Languages

Go, Python (SageMath), C/C++ Rust, LaTeX

Other Activities

Member of CTF Team CodeRed

2020 — 2022

I participated in CTF competitions from time to time, mostly solving crypto challenges.

Developer & Writer of Team Invertible

2020 — Present

I am actively working on Shards of Time, a sokoban puzzle game. We are planning to release the game on Steam.

Other Interests

I love watching films. I've directed several short films, and I'm eager to continue doing so.